SAGMA INTERACTIVE MILLING SUITE

3D Digital Twin & Courseware Library — User & Navigation Guide

Official Technical Operator Manual • Module 2 Section 1.1.1 & Module 4 Mechanical Degermination

PLATFORM VERSION: *2.0 Production*	***ENGINE:*** *Three.js WebGL / WebXR*	***FLEET UNITS:*** *Bucket Elevator & Satake VBF*	***STATUS:*** *Operational / Online*

1. System Overview & Quick Access

The **SAGMA Interactive Milling Suite** is a high-fidelity digital twin and pedagogical simulation environment designed for milling engineers, technical apprentices, and operations personnel. The platform simulates continuous vertical material elevation (Diagram 69b Bucket Elevator) and high-friction maize degermination (Satake VBF Vertical Degermer) with scale-accurate kinematics, internal anatomy inspection, and instant deep-linking to authoritative SAGMA engineering courseware drawings.

ENVIRONMENT	NETWORK / URL ACCESS	RECOMMENDED HARDWARE & BROWSER
Local Development	http://localhost:5173/ (Standard Dev Port)	Google Chrome, MS Edge, Firefox, Safari (Hardware WebGL enabled)
Cloud Deployment	Hosted on Vercel Edge Network (Continuous CI/CD)	Desktop Workstations, Tablets, or Meta Quest / WebXR Immersive Headsets

2. 3D Viewport Controls & Camera Navigation

The 3D canvas utilizes OrbitControls configured specifically for industrial equipment inspection envelopes. You can freely explore the full height of the elevator or the compact decortication rotor of the Satake machine.

ACTION	MOUSE / TRACKPAD INPUT	TOUCHSCREEN GESTURE	DESCRIPTION & BEHAVIOR
Orbit / Rotate	Left-Click + Drag	Single Finger Drag	Rotates camera around the focal center of the active machine (360° horizontal & elevation).
Pan / Move	Right-Click + Drag *(or Shift + Left Drag)*	Two-Finger Drag	Translates the camera plane along X/Y axes to inspect specific heights (head, leg, or boot).
Zoom	Scroll Wheel / Trackpad pinch	Pinch-to-Zoom	Dollies camera smoothly toward or away from the machine envelope (min: 0.8m, max: 25m).
Auto Turntable	Click **Turntable** button	Tap **Turntable** button	Toggles continuous 360° orbit inspection (0.70 rad/s). *Automatically pauses on manual drag!*
Reset View	Click **Reset View** button	Tap **Reset View** button	Re-centers the camera to default framing distance tailored to the active machine's dimensions.
Cinema 16:9	Click **16:9** button	Tap **16:9** button	Toggles immersive wide-aspect ratio mode for training presentations and classroom displays.

3. Equipment Fleet Registry & Machine Profiles

The platform maintains isolated, verified engineering specifications for each machine in the SAGMA fleet. Open the **Fleet Registry** by clicking the Fleet button in the top navigation bar or quick toolbar to switch profiles with a single click.

SAGMA DIAGRAM 69(b) BUCKET ELEVATOR	SATAKE VBF VERTICAL MAIZE DEGERMER
• **Classification:** Vertical Continuous Bulk Elevator • **Structural Height:** 8.55 m continuous leg trunking • **Drive Head:** 391 mm crowned pulley w/ sprag backstop • **Conveying Speed:** Scale-accurate 2.80 m/s linear belt speed • **Tensioning Boot:** Twin-screw take-up with cleanout doors • **Rated Capacity:** 78.5 metric tons per hour • **Safety Sensors:** 4B M300 underspeed & brass thermocouples	• **Classification:** High-Shear Friction Decorticator • **Operating Envelope:** 1.85 m vertical rotor assembly • **Rotor Velocity:** 465 RPM (75 kW main motor drive) • **Capacity:** 2.0–7.0 t/h (VBF 5A: 2–3, 7B: 3–4, 10A: 4–7 t/h) • **Diagram 4(a) Mapping:** 10 Registered Nodes (Node 2.1 to 2.10) • **Decortication Chamber:** Slotted screen cage & dual beater bars • **Aspiration Core:** Hollow-shaft forced air cooling induction • **Load Regulation:** CCU Ammeter automated flap modulation

4. Real-Time Kinematics & Simulation Controls

The digital twin features a live physics-calibrated animation loop that accurately models rotating shafts, circulating belts, and material flow.

CONTROL / TOOL	LOCATION	FUNCTION & OPERATIONAL USE
Start / Stop Machine	Header Quick Toolbar	Starts or halts the mechanical drive loop. When stopped, all motion and acoustic audio cease immediately for static inspection.
Speed Selector	Left Panel / Telemetry	Switch between **0.5x** (slow motion training inspection), **1.0x** (nominal factory operating speed), and **1.5x** (stress test simulation).
Live Telemetry HUD	Header Status Chip	Real-time numeric feedback: Linear Velocity (m/s) or RPM, Motor Electrical Load (A CCU), and Hourly Throughput (t/h).
Casing Opacity (X-Ray)	Floating Controls Bar	Adjustable transparency slider (0% to 100%). Enables looking inside solid steel housings to observe spinning beater bars and bucket travels.
Acoustic Audio FX	Toolbar Speaker Icon	Toggles spatial WebAudio motor hum and conveying sounds, dynamically pitched according to the operating speed.

5. Interactive Anatomy Explorer & 3D HUD Callouts

Operators can inspect machine sub-assemblies through two synchronized interaction paths:

METHOD A: DIRECT 3D MESH INTERACTION	METHOD B: STRUCTURED PARTS REGISTRY DRAWER
1. Move cursor over any component in the 3D viewport. 2. Click directly on the part (e.g., drive pulley or screen cage). 3. The mesh highlights with an emerald selection glow. 4. A **floating 3D HUD banner** projects adjacent to the part displaying its Part #, Name, Section, and Engineering Clearances. 5. The camera dollies smoothly to focus on the selected component.	1. Open the collapsible **Parts Navigation Drawer** on the left. 2. Browse components organized by functional assembly. 3. Click any component card in the list. 4. The 3D viewport automatically frames the part. 5. The lower inspector panel populates full engineering specifications, LOTO safety guidelines, and maintenance intervals.

6. SAGMA Courseware Technical Library

The **Technical Library** modal serves as the authoritative bridge between hands-on 3D simulation and official SAGMA courseware training manuals. It features high-resolution engineering schematics and interactive parts registers for both fleet units.

STEP	ACTION	DETAILS & RESULT
Step 1	Click **Technical Library**	Click the book icon in the top header or navigation rail. The modal opens automatically synced to whichever machine is active in the 3D scene.
Step 2	Select Machine in Header	Use the toggle buttons in the modal header to switch between **Diagram 69(b) Bucket Elevator** and **Satake VBF Maize Degermer**.
Step 3	Browse Drawing Tabs	**Bucket Elevator Tabs:** • *Diagram 69(b) Schematics:* General arrangement & boot/casing dimensions. • *Manual Drawing 19(b):* Head assembly, centrifugal saddle, and drive unit. **Satake VBF Maize Degermer Tabs:** • *Diagram 4(a) Isometric:* General arrangement with magnetic trap & aspiration. • *Drawing 4(a) Exploded:* Detailed exploded rotor, beater bars & screen cage. • *Process Diagram 3:* Mechanical friction decortication process flow.
Step 4	Official Parts Register	Click the **Official Parts Register** tab to inspect the complete schedule of parts (Part #, Component Name, Section, Feed Flow Axis, and Tolerances).
Step 5	**Interactive Deep-Link (Click Row)**	**Click ANY row in the parts table:** The modal automatically closes, switches the 3D scene to that machine (if not active), dollies the camera to the exact component, and displays the 3D HUD banner!

7. Quick Reference & Troubleshooting Cheatsheet

SYMPTOM / QUESTION	CAUSE / EXPLANATION	RECOMMENDED RESOLUTION
Machine motion appears stopped	Simulation loop was paused or stopped via header toggle.	Click the green **Start Machine** button in the top toolbar.
Turntable stops rotating	Turntable automatically disengages whenever user clicks or drags.	Click **Turntable** in the quick toolbar to restart automated 360° rotation.
Cannot see internal components	Solid steel outer casing is at 100% opacity.	Slide the **Casing Opacity** slider down to 25% or 0% (full X-Ray).
Camera is disoriented / too close	Manual zoom or orbit exceeded the optimal framing envelope.	Click **Reset View** to instantly restore the default camera framing.
Switched machine but wrong drawing	Technical Library was opened before switching machine profile.	Use the machine selector buttons at the top of the Library modal.

SOUTHERN AFRICAN GRAIN MILLERS ASSOCIATION (SAGMA)
Vocational Technical Training & Digital Twin Initiative
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INSTRUCTOR SIGN-OFF / DATE:

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Milling Engineering Training Division